

Data Science

1. D

2. B

3. B

4. A

5. A

6. A

7. B

8. C

9. A

10. C

11. B

12. C

13. B

14. C

15. D

16. D

17. B

18. C

19. D

20. C

21. Supervised learning is a type of machine learning algorithm that learn to predict new values based on past observation of a given labeled dataset.

22. Model M1 is the result of ~~not~~ providing ~~enough~~ ^{too much} training data Or not splitting the training and testing data set correctly result in a high accuracy for training while it performed poorly on new unseen data.

25. Major applications of data sciences are

- Recommendation system for streaming platform
- Image search for e-commerce

29. Data normalization is important and required for K-NN because K-NN is a type of lazy machine learning algorithm that ~~doesn't need~~ training. and All the data are mostly classified based on calculation of the distance between each point on the plane.

28. Data pre-processing is the process in a data science life cycle that involve cleaning data, handling missing data and/or transforming data so that the data is reliable for analysis and training.

27. To detect outlier in a data set, which is needed to analyse and produce reliable data we can ~~also~~ calculate the z-score for each data row. which will determine if the data is an outlier depending on the result.

26. Distinguish between Structured and semi-structured data.
- Structured data are data that comes with an orderly manner in the form of rows and columns. For example database, spreadsheet.

- Semi-structured data are data that is structured but in an unordered manner. It can come in different format as well. For example JSON and XML.

